

ChartGenEval: Corruption-Tested Multi-Dimensional Feedback for Rhythm-Game Chart Generation

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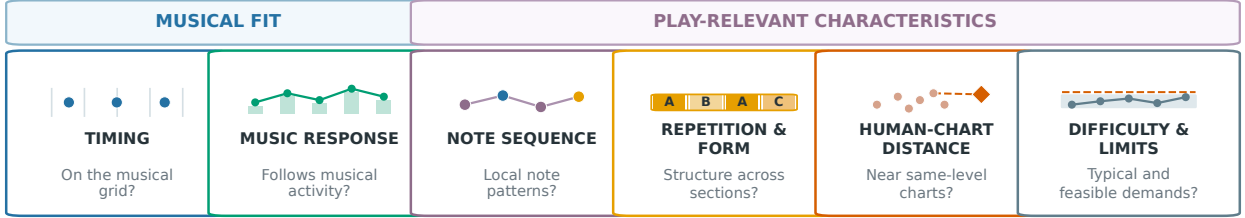


Figure 1: The six dimensions of ChartGenEval. ChartGenEval organizes chart evaluation into six distinct questions, grouped by musical fit and play-relevant characteristics. Each dimension retains its own interpretation.

Abstract

A generated rhythm-game chart need not reproduce one official note sequence: many note choices can fit the same song and difficulty. Reference-note agreement therefore measures reconstruction, not the full design problem. We introduce ChartGenEval, a six-question evaluation framework with an automatic, corruption-tested core. It leaves note choice open while anchoring timing to the song: the matched official chart supplies only its authored timing map, never target notes.

We test each core output with dose-controlled failures rather than assume that a familiar statistic measures chart quality. Across 80 held-out song groups, seven output axes satisfy prespecified sensitivity and invariance criteria in nine nonredundant tests. Complementary stress tests on the 40-song development panel expose two broader lessons. A chart-wide phase estimate recovers injected shifts of 15, 30, and 60 ms while chart-only outputs remain essentially unchanged. Common-pattern rewriting lowers mean language-model perplexity by 37%, and loop collapse raises mean self-similarity by 62%. ChartGenEval therefore reports separate, role-specific signals instead of one proxy or total score. The profile compares measured properties of generator outputs; using these readings as optimization targets or constraints requires additional task-specific validation.

1 Introduction

Chart evaluation is difficult because charting is a design task: multiple timed, typed note sequences can fit the same song and difficulty. Automatic generation has progressed from onset detection to conditional sequence models [1–4], but evaluation must still distinguish valid alternatives from errors.

Agreement with one official chart is therefore useful but incomplete. A placement F1 score measures reconstruction: a valid alternative loses credit in the same way as an error. Other automatic proxies answer narrower questions. Language-model (LM) likelihood measures model fit, while diversity and self-similarity describe generated material. None alone tells a developer which chart property changed or how to improve the generator. Fast model iteration needs automatic feedback that keeps these questions separate.

Our key observation is that freedom of note choice does not remove external constraints. A chart may choose *what* notes to place, but it cannot choose the song’s musical time. We therefore use the matched official chart only as an authored map of bars, meter, and tempo. Its note placements never become targets. This separation preserves alternative designs while exposing chart-wide timing shifts that interval- and type-based statistics cannot see.

ChartGenEval organizes evaluation into the six questions in Figure 1. Its corruption-tested core covers timing, local note sequence, local repetition, and difficulty or playability limits. Each output has a role: a directional signal describes a change, a feasibility check flags a declared limit, and a descriptive value characterizes model behavior. These diagnostic roles do not by themselves establish safe optimization objectives or player-rated quality. The profile keeps them separate.

We test the core outputs by injecting known failures at increasing strengths. A useful output should react to its target failure and remain unchanged under predeclared invariance controls. Seven output axes meet these criteria in nine nonredundant tests on 80 held-out song groups. Complementary development stress tests yield two broader lessons. First, timing must be anchored to the song: our phase estimate recovers 15–60 ms chart-wide shifts that chart-only statistics miss. Second, a plausible proxy can reward the

wrong behavior: common-pattern rewriting improves perplexity, while loop collapse increases self-similarity.

Our contributions are:

1. **Automatic, dimension-specific feedback.** ChartGenEval separates evaluation questions and output roles, so measured properties of generator outputs can be compared without treating one reference note sequence as the only valid answer.
2. **Corruption-tested signals and reusable diagnostic lessons.** Nine nonredundant held-out tests support seven core output axes. Complementary development tests identify a translation blind spot in chart-only timing statistics and wrong-way incentives in perplexity and self-similarity.

2 Related Work

Reference-chart agreement measures reconstruction against one authored sequence. DDC and GenéLive! report F-scores for predicted placements, while TCP adds onset-plus-type F1 and local pattern recall [1, 3, 4]. TaikoNation compares binary timing frames and pattern spaces [2]. These measures remain valuable for reconstruction, but they treat one authored sequence as the answer.

Unpaired statistics and human studies answer complementary questions. DDC uses LM perplexity and token accuracy to measure held-out model fit [1]. Symbolic-music and text-generation work adds diversity, distribution, self-similarity, and repetition statistics [5–9]. Human and expert studies instead evaluate system-level experience [3, 10, 11]. ChartGenEval complements these approaches by testing whether automatic outputs respond to specified chart failures.

Metric stress tests show that a plausible statistic can miss a musical failure. SCHmUBERT gives a proof-of-concept piano roll with nearly indistinguishable framewise self-similarity statistics from its source, despite poor musical quality [12]. Fréchet Music Distance is tested with pitch and velocity perturbations, but remains a set-level embedding distance [13]. STRUM reports reference-onset F1 after a per-song global-offset search [14]; ChartGenEval reports the offset itself without using reference notes as targets. Its controlled corruptions test both target response and invariance, turning metric choice into an empirical question.

3 Methods

3.1 Evaluation questions and core outputs

ChartGenEval treats the six dimensions in Figure 1 as separate questions. The main analysis focuses on seven outputs for which we have held-out controlled-corruption evidence (Table 1). Timing uses three readings because dense noise, rare outliers, and a shared offset are different failures. The other outputs cover local transitions, local repetition, requested note rate, and density spikes.

All outputs are automatic, but automation does not make them interchangeable. Grid-phase error and timing tails

have a clear direction; density spikes provide a course-relative density constraint; transition familiarity and typicality describe local patterns without establishing their quality. Raw LM loss is retained only as a description of model fit and as a stress-test baseline. In particular, an unseen transition is unfamiliar to the training corpus, not necessarily invalid.

3.2 Same-course calibration

Several chart properties have no universal “more is better” direction. A chart can be too sparse or too dense, and too little or too much repetition can both be undesirable. We therefore measure departure from the middle 80% of same-course human charts. For a feature value x , let l and u be the 10th and 90th training percentiles and $h = \max((u - l)/2, 10^{-9})$. The band score is

$$d(x) = \begin{cases} (l - x)/h, & x < l, \\ 0, & l \leq x \leq u, \\ (x - u)/h, & x > u, \end{cases} \quad (1)$$

$$s(x) = \frac{1}{1 + d(x)^2}.$$

Thus $s(x) = 1$ inside the band and decreases symmetrically outside it. This is a course-conditioned typicality reading, not a verdict on creativity or overall quality. One-sided limits, such as the density-spike check, remain separate because a severe local overload should not be canceled by another high score.

The three less familiar outputs use simple transformations. If r is the rate of interval-type trigrams unseen in same-course training charts, transition familiarity is $\exp(-r/0.08)$. For M 4-gram windows with U unique windows, the raw repetition rate is $(M - U)/M$ before band calibration. The density-spike score is $\exp(-e/1.5)$, where e is the amount by which a chart’s 95th-percentile local density jump exceeds the same-course limit.

3.3 Data and timing anchors

We separated design from confirmation. The seven core outputs and their corruptions were designed on 40 development song groups with 170 charts, then tested on 80 held-out groups containing 333 charts. The calibration set contains 3,880 course charts from 813 separate song groups. Songs, rather than courses or corruption replicates, are the inferential units. Appendix A gives the corpus construction, completeness rules, and full analysis protocol.

Timing requires a reference outside the generated event sequence. Our primary grid comes from the matched official file’s authored bar timestamps, meter, and tempo. We never use its note times or note types as targets, and we never accept the grid declared by the generated chart. The first choice preserves freedom of note design; the second prevents a generator from redefining the clock against which it is evaluated. An audio-estimated alternative is reported in Appendix A.

Evaluation question	Automatic reading	Present evidence
Timing	Timing clean rate, excess-error p99, and whole-chart grid-phase offset	C1, C1s, and C2 held-out tests
Note sequence	Transition familiarity: a one-sided score derived from the rate of unseen interval-type trigrams	C3 held-out test
Local repetition; long-form structure	Same-course typicality of repeated versus unique 4-grams	C4, C5, and C8 test local repetition; long-form readings remain exploratory
Response to music	Bar-scale energy response and onset support	Exploratory development analysis; Appendix C
Human-chart distance	Distance from the same-course training distribution	Exploratory development analysis; Appendix C
Difficulty and limits	Same-course note-rate typicality and a local density-spike limit	C6 and C7 held-out tests

Table 1: Evidence differs across the six questions: seven core outputs have held-out tests, while the response-to-music, human-distance, and long-form readings remain exploratory.

3.4 Measuring timing against the song

A common shift δ leaves every inter-note interval unchanged:

$$(t_{i+1} + \delta) - (t_i + \delta) = t_{i+1} - t_i. \quad (2)$$

The note types are also unchanged. Chart-only measurements built from these intervals and types therefore cannot identify the shift. This translation symmetry explains the blind spot before any experiment is run.

Event matching and timing quality serve different purposes. Onset evaluation first decides whether an event was recovered: `mir_eval` defaults to a ± 50 ms matching window [15], while chart-generation evaluations use, for example, ± 20 ms in DDC and ± 50 ms in GenéLive! [1, 3]. These windows permit event association despite timing uncertainty. For fine-grained chart timing, however, deviations inside such a window still matter: matching credit alone does not describe how precisely a note follows the musical grid. We therefore separate assignment from the error scored after assignment.

For a note at t_i assigned to an authored-grid anchor g_i , let $e_i = t_i - g_i$ be its signed displacement and $[z]_+ = \max(z, 0)$. We report raw offsets separately from the tolerance-subtracted error

$$a_i = [|e_i| - \tau_{\text{abs}}]_+, \quad \tau_{\text{abs}} = 6 \text{ ms}. \quad (3)$$

The tolerance band assigns zero error to small deviations and measures only their excess beyond the chosen perceptual scale. Friberg and Sundberg [16] report an approximately 6 ms just-noticeable difference for a single displaced tone in short isochronous sequences. This motivates a stringent operating threshold, not a universal claim that all sub-6 ms deviations in gameplay are inaudible. Its purpose is finer timing assessment than event-recovery credit; a smaller threshold alone does not establish greater perceptual validity.

Timing clean rate is the fraction of all notes matched within this band. The *excess-error p99* is the within-chart 99th percentile of a_i over matched notes, not the percentile of $|e_i|$ or the injected corruption magnitude. Unmatched notes remain in the clean-rate denominator and are reported separately. This tail reading complements the clean rate, which must also be read with note rate. Appendix A separates assignment gates from the 6/12/18 ms reporting bands.

Relative timing measures interval distortion. For consecutive matched notes with increasing anchor times, define $\Delta_i = g_{i+1} - g_i > 0$ and

$$\begin{aligned} r_i &= |(t_{i+1} - t_i) - \Delta_i| = |e_{i+1} - e_i|, \\ \theta_i &= \max(6 \text{ ms}, 0.025\Delta_i), \quad q_i = [r_i - \theta_i]_+. \end{aligned} \quad (4)$$

The envelope follows the approximately 6 ms discrimination floor below 240 ms and 2.5% relative threshold for longer intervals in the same psychophysical study [16]. A shared displacement cancels in r_i , whereas opposite-signed neighboring offsets can compound. For example, offsets of +4 and -4 ms across a 200 ms anchor interval produce 8 ms of interval distortion and 2 ms of excess, even though each absolute offset lies within the 6 ms band. Absolute offset, relative distortion, and shared phase thus expose different timing failures. Relative summaries are complementary profile readings; the held-out C1s target is excess-error p99 from Equation 3.

Nearest-grid matching can miss a constant shift (Figure 2). On a dense grid, each shifted note may simply be paired with another nearby grid point. We therefore estimate one offset for the whole chart. Let G_k be the fixed bar, beat, and eighth-note grids, and let $d(x, G_k) = \min_{g \in G_k} |x - g|$. For note times t_i , we score a candidate shift by

$$\begin{aligned} S(\delta) &= \sum_k w_k \frac{1}{n} \sum_i \left[1 - \frac{d(t_i - \delta, G_k)}{\tau} \right]_+, \\ \hat{\delta} &= \arg \max_{\delta \in [-250, 250] \text{ ms}} S(\delta), \end{aligned} \quad (5)$$

In Equation 5, $\tau = 12$ ms, and the bar, beat, and eighth-note weights are 0.2, 0.3, and 0.5. We search in 0.5 ms steps. Combining three grid levels reduces periodic ties. The resulting $\hat{\delta}$ is the selected chart-level offset within the search window, not a guarantee of unique alignment: periodic aliases, near ties, and syncopated content can remain ambiguous. Recovery is established here only for the tested 15, 30, and 60 ms shifts. BPM alone provides a period but not a phase, so it cannot replace this anchor.

3.5 Controlled corruptions and acceptance rules

Starting from human charts, each corruption strengthens one operational failure. Other outputs may also move, so

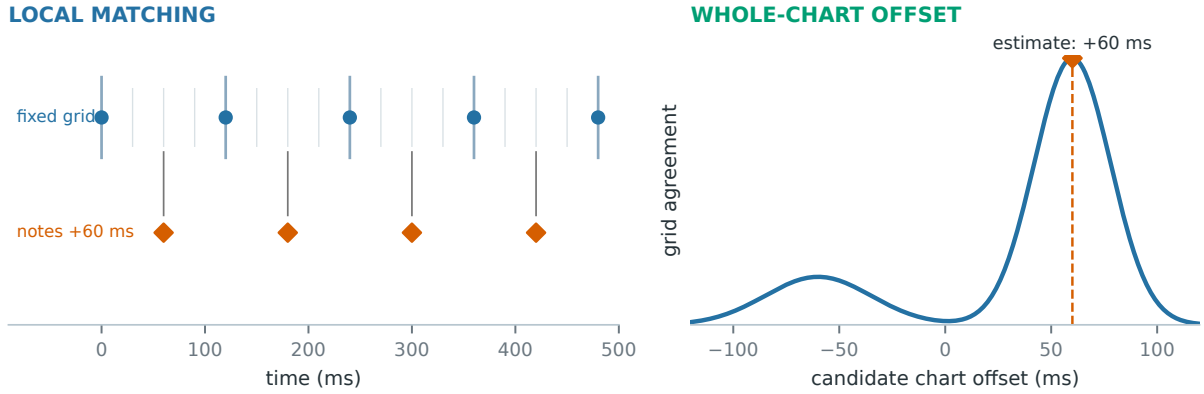


Figure 2: Why timing needs a whole-chart estimate. **Left:** nearest-grid matching can pair shifted notes with different subdivision points and report small local errors. **Right:** a schematic agreement curve over candidate whole-chart offsets; the curve peaks at the injected +60 ms shift. Local plausibility does not imply correct song alignment.

the tests establish target sensitivity rather than exclusive specificity. The target output and expected direction were fixed before the held-out run; all transformations appear in Appendix B.

After orienting outputs so larger is better, support requires a negative dose-rank association, a negative strongest-dose contrast against the matched control, and every relevant invariance check to pass. We use song-cluster bootstrap intervals adjusted across ten prespecified rows. Full decision rules and sample accounting appear in Appendix A.

4 Results

4.1 Core outputs respond to target failures

Every core output has the expected negative dose-rank association under its target failures (Table 2). The result covers nine nonredundant corruption–measurement pairs but seven output axes because the same 4-gram axis detects three distinct edits. All applicable invariance controls pass. The important result is not that every corruption needs a new metric. A small set of interpretable outputs can cover different failure mechanisms when each output is tested against a concrete claim.

The sparse-error result measures excess beyond the 6 ms band after grid assignment; its strongest-dose contrast is only 0.306 ms on the larger-is-worse excess-error scale. It supports sensitivity under this protocol, not recovery of the injected 60 ms displacement. Negative dose-rank association also need not imply strict monotonicity: C1 mean clean rates are 0.996, 0.616, 0.408, and 0.495 from intact to strongest jitter. The highest-dose rebound is consistent with grid re-matching, although these data do not isolate its cause. The clean rate is therefore a threshold reading, not a monotonic severity loss. Complete sample counts and exclusions appear in Appendix A.

4.2 External time detects chart–audio shifts

Translation symmetry predicts that a chart can look internally plausible yet be late relative to the song. The 40-song

development experiment confirms this mechanism. After shifting every note by 15, 30, or 60 ms, no primary chart-only output changes by more than 0.03 human standard deviations. The phase estimator’s median matches each injected shift. With either the authored or audio-estimated grid, at least 169 of 170 charts are within 3 ms of the injected value.

Local matching and global phase answer different questions. A dense grid can re-pair a shifted note to another subdivision and still report a small local error. The whole-chart estimate preserves the shared offset instead. The same principle also limits local tail summaries: their values depend on the selected grid assignments and tolerance band. Timing evaluation therefore keeps raw offsets, excess errors, relative intervals, and shared phase separate, so tolerance-subtracted values are not mistaken for absolute displacements.

4.3 Common proxies can reward the wrong behavior

Two familiar proxies move in the wrong direction under targeted failures (Figure 3). LM perplexity rewards common-pattern rewriting: on the development panel, mean perplexity falls by 37%, from 9.44 to 5.98. A lower-is-better reading therefore prefers the corrupted chart, opposite to the intended property [17]. The reversal remains with a separately fitted scoring LM, while 4-gram typicality moves away from the human range. Secondary checks appear in Appendix C.

If self-similarity is maximized as a quality objective, it rewards loop collapse. Repeating the dominant 4-gram raises mean self-similarity by 62% even though the note-type stream has collapsed into a four-event cycle [9]. 4-gram typicality moves in the opposite direction. The practical rule is to stress-test a candidate reward against the failure it is meant to prevent.

Automatic profiles expose which measured property changed during model development. Phase and excess-error summaries quantify distinct timing deviations; density spikes provide a course-relative constraint; familiarity

Targeted failure	Core output	Dose trend [99.5% CI]	Strongest-dose change [99.5% CI]
Dense timing jitter	timing clean rate	-.744 [-.815, -.667]	-.502 [-.558, -.446]
Sparse 60 ms timing errors	excess-error p99	-.234 [-.344, -.134]	-.306 [-.508, -.144] ms
Whole-chart shift	grid-phase offset	-.916 [-.981, -.826]	-55.58 [-61.43, -49.83] ms
Note-type shuffle	transition familiarity	-.800 [-.856, -.737]	-.231 [-.262, -.199]
Loop collapse	4-gram typicality	-.445 [-.573, -.312]	-.165 [-.226, -.108]
Common-pattern rewrite	4-gram typicality	-.358 [-.459, -.258]	-.131 [-.189, -.078]
Note-rate scaling	note-rate typicality	-.693 [-.784, -.579]	-.582 [-.672, -.483]
Local burst insertion	density-spike limit	-.940 [-.950, -.930]	-.619 [-.644, -.593]
Fixed-window shuffle	4-gram typicality	-.589 [-.728, -.435]	-.224 [-.293, -.156]

Table 2: Held-out results on 80 song groups. Outputs are oriented so larger is better; a useful response is therefore negative as a failure becomes stronger. Every row meets both criteria, and all applicable controls pass. Individual 99.5% song-cluster bootstrap intervals use a Bonferroni adjustment across ten prespecified rows; both endpoints must pass to support a row, controlling family-wise error at at most 5%. The two originally named 4-gram variety and repetition scores are algebraic mirrors, so they appear once as one effective axis.

and typicality describe local patterns. Using these readings as optimization targets requires additional task-specific stress testing, including the grid-aliasing behavior above. The present results establish responses to specified edits, not improvement in player judgments or optimization outcomes.

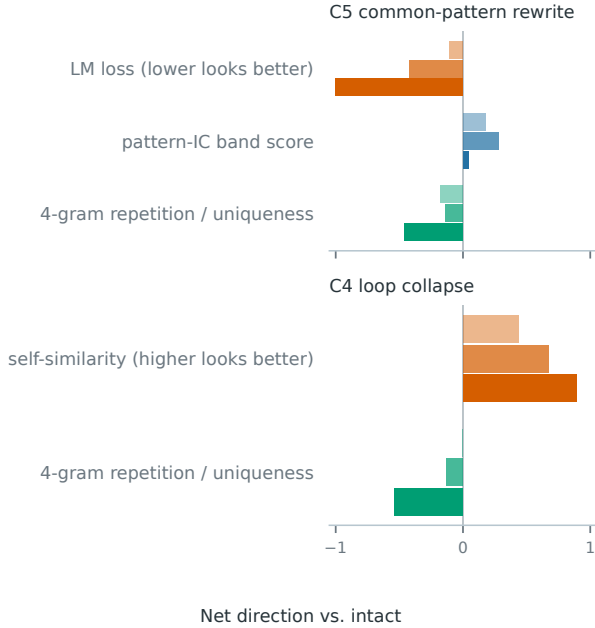


Figure 3: Wrong-way incentives on the development panel. Common-pattern rewriting improves LM loss and its band score; loop collapse raises self-similarity. In both cases, 4-gram typicality falls. Bars show net direction from the intact chart; darker bars mark stronger corruption.

5 Discussion

Typicality also needs context. A corruption often pushes a chart outside the same-course human band, but the reverse implication is invalid: an unusual chart may be deliberate and playable. Separate timing and feasibility checks help distinguish novelty from a concrete failure. When signals disagree, the disagreement is information rather than an error to hide with an average.

The descriptive public-system profiles in Appendix C.3 illustrate two reusable reading habits. Similar clean-note fractions can hide different timing tails and global offsets, and a generator can match one difficulty while overshooting another. These examples are not additional metric validation; they show why diagnosis should follow failure mechanisms rather than one headline score.

6 Limitations

Controlled corruptions establish response to defined failures, not a universal ranking of chart quality. The present set does not cover every error, such as wrong-song pairing or removal of musically important notes. The response-to-music, long-form, and human-distance extensions were designed on the development panel and remain exploratory.

The human bands come from one Taiko corpus. Song-cluster intervals assume exchangeability across songs; shared authors or sources across song groups are not separately modeled. Timing also requires an external grid; without an authored map or a reliable audio estimate, the timing outputs are unavailable. Finally, timing clean rate has no reference-note recall term, so deleting off-grid notes can improve it. This is why the profile reads clean rate together with note rate and keeps feasibility checks separate.

7 Conclusion

ChartGenEval turns chart evaluation into separate, testable feedback signals. Seven output axes pass nine held-out controlled-corruption tests. On the development panel, the song-anchored phase estimate recovers injected shifts invisible to chart-only statistics. Those development tests also show that likelihood and self-similarity can move in the wrong direction under targeted failures, so neither should be used alone as a quality score. The broader lesson is simple: automatic evaluation is most useful when each signal names a failure, survives a targeted stress test, and keeps its role visible to the model developer.

Compliance with Ethical Standards

The study uses community-transcribed charts and associated audio gathered from online sources as described in

Section 3.3; these potentially copyrighted materials are not redistributed. The released materials exclude source charts, audio, personal data, and generators. Automated evaluation can lower the cost of content production, so high-volume deployment still requires platform policy and moderation. This study analyzes chart content, audio, and derived numerical records; it does not recruit participants or use player profiles, gameplay records, or interventions.

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A Corpus Coverage and Held-Out Evaluation Protocol

A.1 Corpus coverage

The calibration corpus is summarized by course, star level, and tempo. Figure 4 shows its difficulty and tempo coverage.

A.2 Corpus and timing details

We use Taiko-style chart files and associated audio gathered from community resources available online. The training split contains 924 source rows that resolve to 813 unique song groups and 3,880 course charts. The test split contains 140 source rows that resolve to 120 song groups. The underlying charts and audio may be copyrighted, so they are not redistributed.

The primary timing grid uses authored per-bar timestamps, meter, and tempo. Notes are assigned one-to-one to grid anchors by minimum-cost matching. Candidate gates depend on anchor spacing and are bounded between 10 and 50 ms; these assignment gates are distinct from the 6 ms error dead zone. For local timing diagnostics, every generated note is classified as within 6 ms, in the 6–12, 12–18, or above-18 ms range, or unmatched. We also compare adjacent intervals with tolerance $\max(6 \text{ ms}, 2.5\% \times \text{IOI})$, where IOI is the interval between consecutive matched anchors. The public fields `absolute_error_p99_ms` and `relative_error_p99_ms` summarize the excess errors a_i and q_i ; raw absolute offsets and raw interval distortions are retained in separate fields. Percentiles are computed within each chart before the replicate, course, and song averages described below. An audio-downbeat estimate provides a development-stage alternative: its period is within 2% of the metadata period on 39 of 40 development songs and differs by 2.39% on the remaining song. Held-out absolute offsets use the authored grid because it fixes bar phase directly.

A.3 Panel construction and integrity

Source rows are canonicalized by `group_id` and audio SHA-256; inconsistent aliases are rejected before panel indexing. The test split contains 120 canonical song groups in stored order. Groups 0–39 form the development panel, and groups 40–119 form the held-out panel. Before inspecting held-out results, we fixed the dataset revision, corruptions and doses, target measurements, controls, thresholds, calibration, and analysis plan.

A.4 Replicates, completeness, and inferential units

Every nonzero corruption cell and control condition uses five fixed base seeds. Replicates are averaged within chart and dose. A course is complete only when its intact value, every target dose and replicate, matched control, and required checks are available. A song enters a test with at least one complete course, and complete courses are averaged within song. Songs, rather than charts, courses, or replicates, are the inferential units for both response statistics. Deterministic no-op and missing cells do not enter the analysis.

A.5 Control checks and numerical invariance

Identity re-evaluation applies to every retained measurement. A joint +1-second shift of notes, authored grid, and duration tests invariance to the time origin. A Don–Ka color bijection applies to timing and density, where color is irrelevant; it is not a grammar or form control. The 60 ms C2 translation is the matched control for chart-only density, grammar, and structure measurements. A required control failure prevents support for the affected target response. Each measurement uses a specified equivalence tolerance and timestamp-rounding rule. Chart-only interval arithmetic uses integer microseconds, three orders of magnitude finer than the smallest reported tolerance.

A.6 Intervals and decision rule

After orienting outputs so that larger is better, the first statistic is the within-chart Spearman correlation between the four severities (intact plus three doses) and the measurement; an all-constant response has zero sensitivity. The second statistic is the oriented maximum-dose value minus the matched control. Point estimates average song-level values. We use 10,000 song-cluster bootstrap draws: secondary analyses report nominal 95% intervals, and the ten primary rows use Bonferroni-adjusted 99.5% intervals. C6 assigns severities $[0, 0.5, 0.5, 1]$ to the intact, 0.5 $\times$, 1.5 $\times$, and 2 $\times$ conditions, so equal-magnitude low and high departures are tied; tied severities receive average ranks. A response is supported when at least 72 of 80 songs are evaluable, all controls pass, and both adjusted upper bounds are below zero. A nonnegative adjusted lower bound contradicts the response; other cases are inconclusive. The two C5 output rows were required to meet the rule together, although their algebraic equivalence leaves one effective

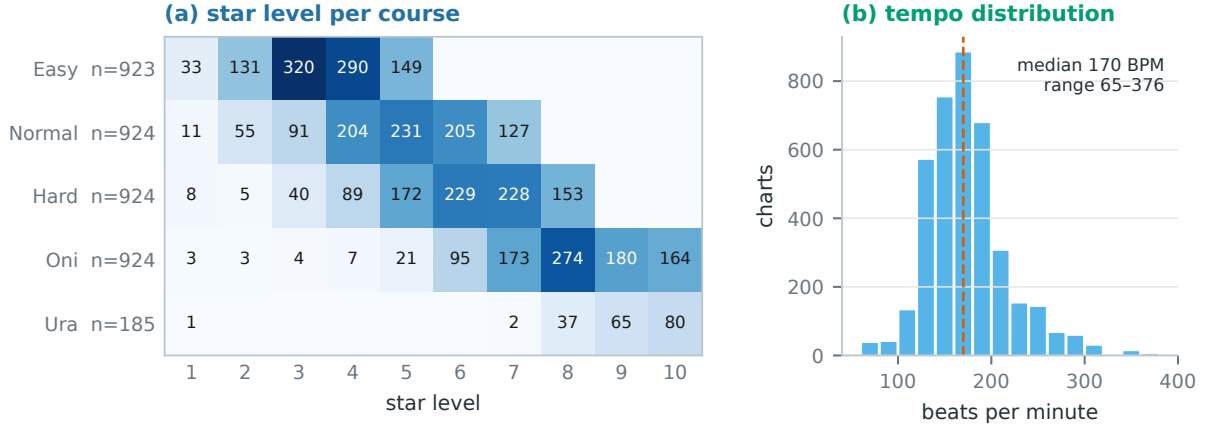


Figure 4: Course difficulty and tempo coverage of the human reference corpus that defines the calibration bands. **(a)** Star-level counts per course across the 3,880 training charts: each course spans a broad, shifting difficulty range. **(b)** Tempo histogram (median 170 BPM, range 65–376).

axis. The C5 LM uses the full stored-order training split, trigram order 3, and $\text{add-}\alpha = 0.05$.

LM scores them. The perplexity reversal remains ($9.44 \rightarrow 6.04$, -36%).

A.7 Sample counts and exclusions

The held-out execution produced 50,283 observation records from 333 charts. C1s retained 332 complete courses and C8 retained 330; every other effective pair retained 333. All 32 controls passed with maximum observed absolute difference zero. Nineteen deterministic no-op cells were excluded.

B Controlled-Corruption Specifications

Figure 5 visualizes the operational change made by each corruption and its prespecified target measurement.

C8 deliberately uses fixed-duration windows so the operation can be applied when authored bar boundaries are unavailable. For a chart-level tempo B in beats per minute, the window length is $L = 4 \cdot 60/B$ seconds, with origin at the first note. Each occupied window is selected with probability p ; the selected windows are permuted, preserving note types and offsets within each window. The four-beat nominal duration sets a common analysis scale, not a claim of alignment to musical bars: meter changes, tempo changes, and pickups need not coincide with its boundaries. C8 tests sensitivity to reordering fixed windows and the resulting event transitions; musical-form validity requires separate evidence. The frozen records retain the identifier `C8_bar_shuffle` for this unchanged operator.

The 170-chart development matrix uses one realization per nonzero cell, producing 4,760 variants including the originals. C1s moves $\max(1, \text{round}(np))$ of a chart’s n notes by ± 60 ms. C3–C5 preserve note times, while C4 and C6 preserve or control density. Raw features and their derived scores are evaluated on the same variants; equivalent quantities are counted once. For C5, one LM chooses common-pattern rewrites and a separately fitted

Controlled corruption (strength)	Operational change	Prespecified response
C1 timing jitter ($\pm 10/20/30$ ms)	all note times	timing clean rate $\downarrow$
C1s sparse timing errors (0.5/1/2%, ± 60 ms)	a few note times	excess-error p99 $\uparrow$
C2 whole-chart shift ($\pm 15/30/60$ ms)	chart-song alignment	grid-phase offset $\uparrow$
C3 note-type shuffle ($p=0.2/0.4/0.8$)	local transitions	transition familiarity $\downarrow$
C4 loop collapse ($p=0.3/0.6/1$)	local repetition	4-gram typicality $\downarrow$
C5 common-pattern rewrite ($p=0.3/0.6/1$)	pattern variety	4-gram typicality $\downarrow$
C6 note-rate scaling ($\times 0.5/1.5/2$)	difficulty fit	note-rate typicality $\downarrow$
C7 burst insertion (2/4/8 clusters)	local overload	density-spike limit $\downarrow$
C8 fixed-window shuffle ($p=0.3/0.6/1$)	fixed-window ordering	4-gram typicality $\downarrow$

Table 3: Controlled corruptions, strengths, and target readings. The edits target operational failures rather than claim to exhaust chart quality. C3–C5 preserve note times; C4 and C6 preserve or control density.

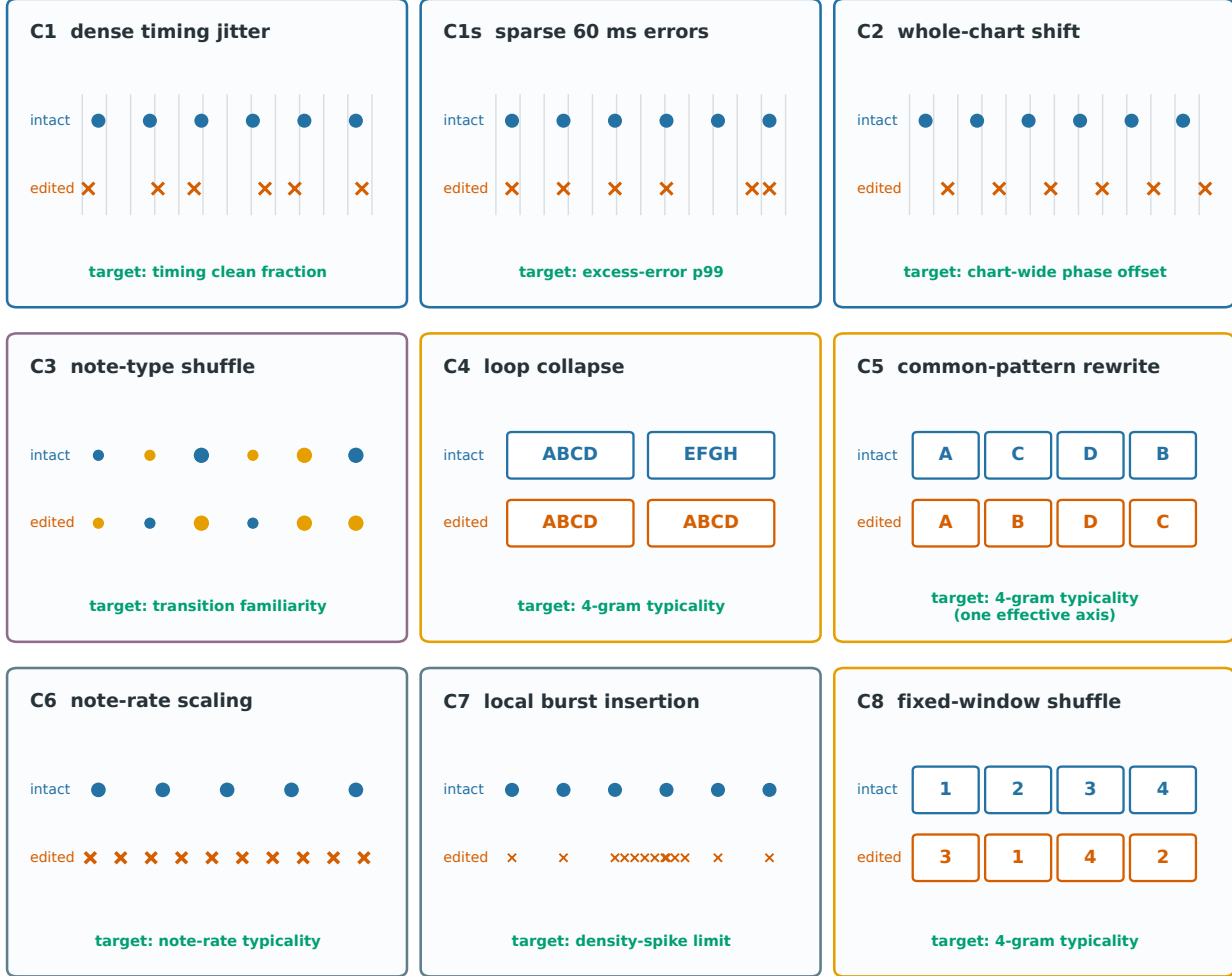


Figure 5: Controlled-corruption atlas. Blue marks the intact chart and orange marks edited events. Each panel shows the property targeted by one corruption and names the target measurement. C1–C2 target timing, C3 targets transition familiarity, C4–C5 target local repetition, C8 rearranges fixed-duration windows, and C6–C7 target difficulty fit and local load. These edits do not exhaust chart quality.

C Metric Guide and Extended Results

C.1 Exploratory measurements

Five additional components were selected or revised on the 40-song development panel. They remain useful hypotheses, but the main paper does not treat them as held-out-confirmed targets. *Reciprocity* compares two-bar phrases

and rewards material that returns after an intervening phrase without rewarding adjacent copying. *Stagnation-alienation* marks active bars in long repeated runs or without a related bar within a 32-bar neighborhood. *Density-energy response* is the rank correlation between bar-level note density and mel energy. *Run-head onset support* asks whether the first note of a rhythmic group lies near strong spectral flux. *Human-chart distance* measures the mean

distance to the 20 nearest same-course training charts in a standardized 32-feature representation.

Their development responses clarify their scope. Reciprocity reacts to loop collapse and fixed-window shuffling; stagnation–alienation reacts most strongly to loop collapse. Run-head support detects whole-chart shifts, while density–energy response detects reordered fixed windows but is weak for short bursts. Human-chart distance detects many distributional departures but does not say whether an unusual chart is poor or creative. These outputs are therefore reported as exploratory monitors rather than core rewards.

C.2 Proxy diagnostics

The common-pattern rewrite is not an artifact of using the same LM for editing and scoring. When one LM selects the rewrite and a separately fitted LM scores it, perplexity still falls from 9.44 to 6.04 (−36%). A symmetric band transform of LM loss also fails to identify the maximum rewrite dose: its held-out change is +0.021 with a nominal 95% interval of $[-0.034, 0.077]$. The 4-gram axis supplies the missing reading. Under loop collapse, set-level Self-BLEU rises from 0.845 to 0.898, also recording the loss of diversity.

Figure 6 connects each construct to its required inputs, representative outputs, and strongest present evidence. The evidence labels distinguish outputs with held-out corruption tests from components supported by development analyses.

Figure 7 shows that targeted corruptions can affect multiple score families, motivating interpretation as a profile rather than as isolated interchangeable measurements.

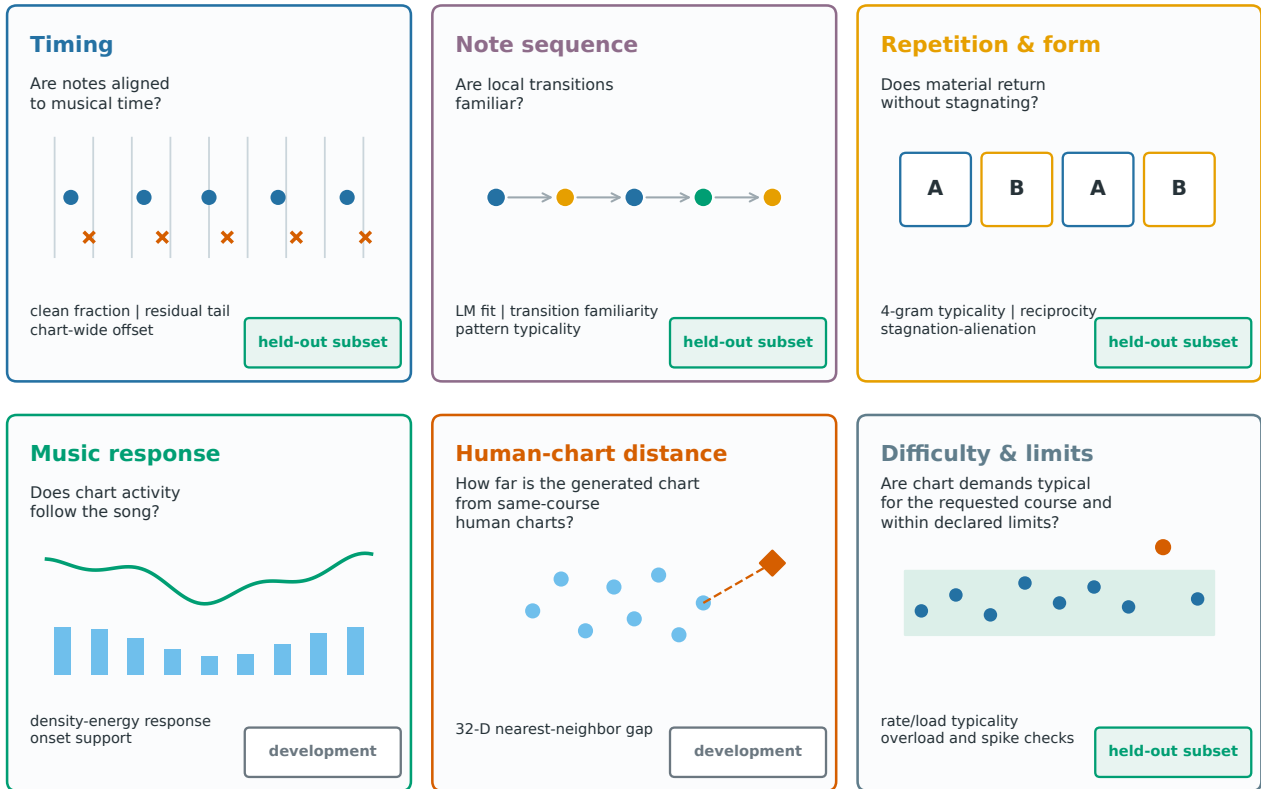


Figure 6: Metric-to-construct guide. Each panel states an evaluation question, illustrates the relevant signal, lists representative outputs, and marks the strongest evidence. “Held-out subset” means that only the named outputs in Table 1 have held-out corruption evidence. Typicality describes departure from same-course human ranges.

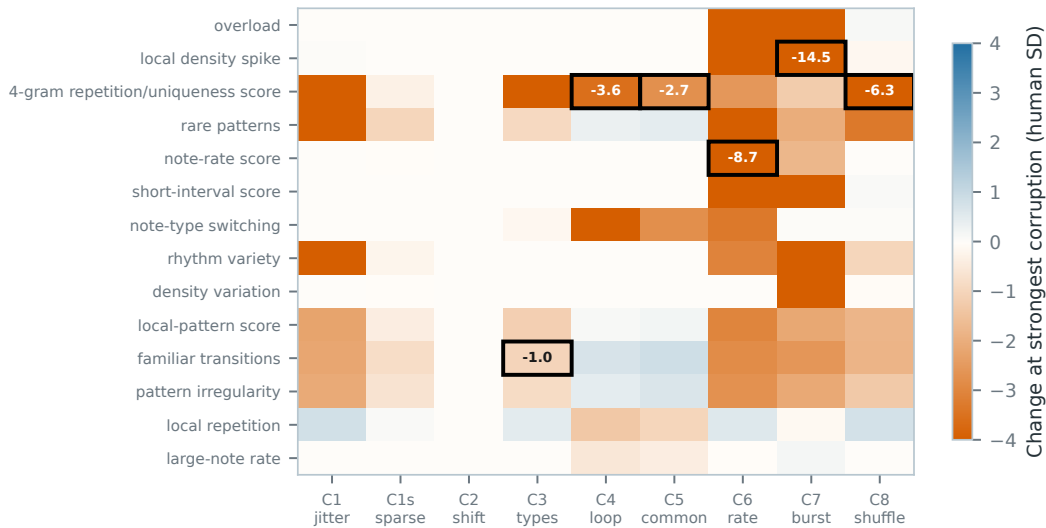


Figure 7: How 14 nonredundant calibrated chart-only scores respond to each corruption at maximum strength on the development panel. Cells cover band-typicality scores and one-sided checks. Red means the score falls and blue that it rises; values are in standard deviations of the human reference charts, clipped at ± 4 . Black outlines mark the target rows selected on the development panel; they do not mark statistical significance. The matrix shows both target sensitivity and cross-family coupling.

C.3 Public-system profiles

We apply the profile to five public generators on the 40 development songs. Mapperatorinator v32 [18] and TaikoNation [2] produce Taiko note types. DDC’s onset pipeline [1], the public timing configuration of GenéLive! [3], and AutoOsu [19] enter only the applicable timing and audio-response readings. We use their public inference configurations without training or fine-tuning on this corpus. These profiles illustrate interpretation; they are not held-out metric tests or a generator leaderboard.

The two Taiko generators have similar median clean-note fractions, yet their relative-interval errors and global phase offsets differ. This is the same local-versus-global distinction exposed by the shift corruption. Difficulty conditioning reveals another pattern: Mapperatorinator overshoots human note rates most on easier courses, while its timing profile changes little by course. The useful conclusion is a course mismatch, not a general claim that the system improves at higher difficulty.

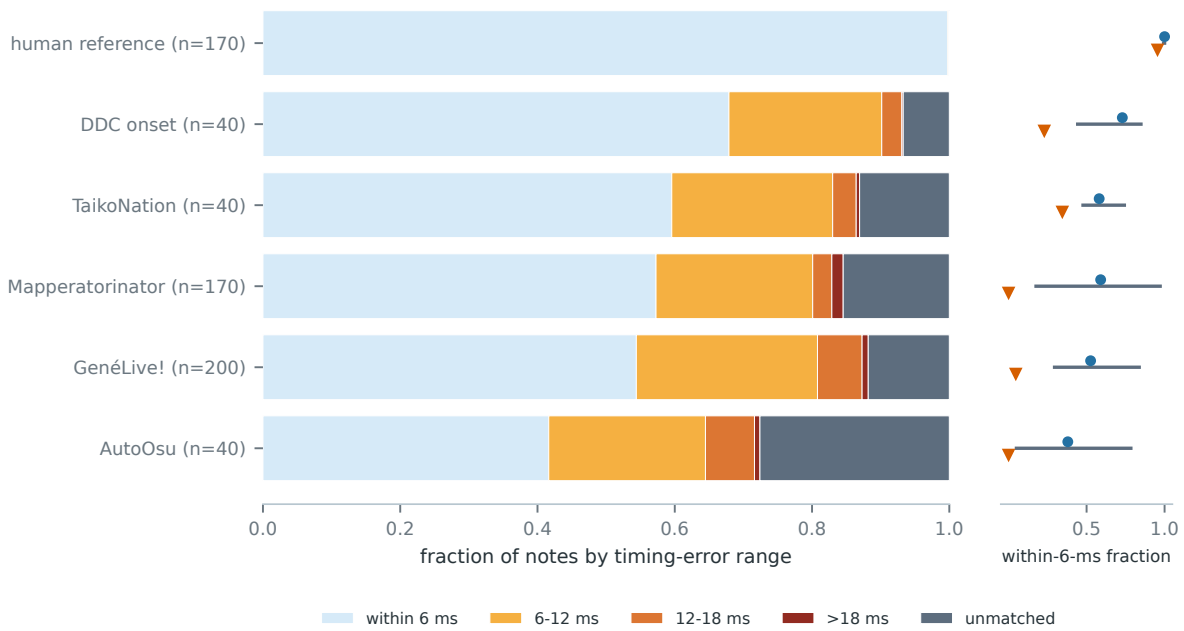


Figure 8: Timing profiles on the 40 development songs. The stacked bars separate locally clean notes from timing tails and unmatched notes; the right strip shows per-chart variation. Systems with similar mean clean fractions can have different tails, so one timing average does not identify the failure.

	Timing within 6 ms	Note rate	Short intervals	Familiar transitions	Local patterns	4-gram repetition / uniqueness	overload	density spike	playability
human reference	1.00	1.00	1.00	0.75	1.00	1.00	✓	✓	✓
Mapperatorinator	0.59	0.23	0.36	0.23	0.19	1.00	✓	✓	✗
TaikoNation	0.58	1.00	1.00	0.00	0.01	0.08	✓	✓	✓

Figure 9: Separate development readings for human charts and the two generators that produce Taiko note types. Columns retain their own meanings; the matrix is a diagnostic profile, not a total score.

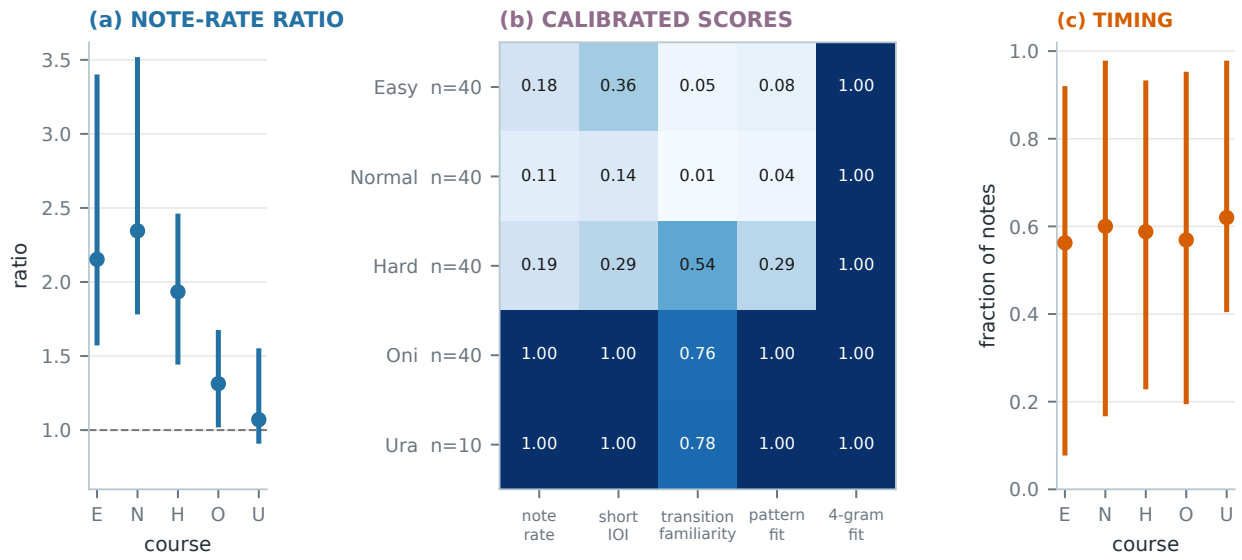


Figure 10: Exploratory difficulty profile for Mapperatorinator on paired development charts. Easy through Oni contain 40 song–course pairs; Ura contains 10. Dots are medians and vertical lines span the 10th to 90th percentiles. Generated density approaches the human reference as course difficulty rises, while timing does not follow the same trend; this supports a course mismatch rather than general improvement at high difficulty.

D Released Materials

Current code, released evaluation summaries and records, and plotting scripts are available at <https://github.com/JacobLinCool/ChartGenEval>. The confirmatory release binds provenance through hashes, but does not package the exact dirty working-tree source snapshot at freeze time or all raw confirmatory records. It therefore supports summary-level auditability, not independent end-to-end reproduction of the confirmatory run from the released materials alone. No source charts or audio are distributed.

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